

# NAB2 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes NAB2 gene fusions from the msk\_impact\_50k\_2026 study, covering 72 fusion events. 25/72 fusions (34.7%) were in-frame. 49/72 fusions (68.1%) retained the PF04905 domain. Domain retention was tested with Fisher's exact test ( $p=0.11258$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for NAB2.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 72 analyzed fusion events, identifying 14 distinct fusion partner genes. The most frequent partner was STAT6, observed in 53 events.

### Domain retention

PF04905 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 20/25 (80.0%) of in-frame fusions retained the domain,  $p=0.11258$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Domain-disruption analysis was skipped: NAB2 has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 71 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 372 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is 6 aa from the nearest configured domain boundary at 366 aa.

### Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NAB2 has no group\_field configured; confidence-stats analysis was skipped.

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

## Joint partner

Joint partner reported: NAB2 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|---------------------|------|
| STAT6        | 53          | 0.7361111111111112   | 0.20043213737826426    | None                    | 0.3434207302131831       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.460649253156967   | 1    |
| CUX2         | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.948051948051948        | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2806892760271785  | 2    |
| PDZRN4       | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.948051948051948        | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2806892760271785  | 3    |
| MGAT4C       | 2           | 0.027777777777777776 | 0.20043213737826426    | None                    | 0.9090909090909091       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.27829286291647964 | 4    |
| KRT6A        | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.935064935064935        | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2779063446728186  | 5    |
| LRP1         | 3           | 0.041666666666666666 | 0.20043213737826426    | None                    | 0.8658008658008658       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2749688060209942  | 6    |
| TMEM117      | 2           | 0.027777777777777776 | 0.20043213737826426    | None                    | 0.6558441558441559       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.22402570150646114 | 7    |
| KIF5A        | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.6623376623376623       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21946478623126012 | 8    |
| FMN1         | 3           | 0.041666666666666666 | 0.20043213737826426    | None                    | 0.23376623376623373      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.13953281344214447 | 9    |
| DLGAP2       | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.23376623376623373      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.12762805153738258 | 10   |
| FAM186A      | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.23376623376623373      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.12762805153738258 | 11   |
| LFNG         | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | None                     | None                       | ['domain_retention', 'recurrence']                       | 0.09868127456587769 | 12   |
| LGR5         | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.09740259740259738      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.09840727231660334 | 13   |
| MTOR         | 1           | 0.013888888888888888 | 0.20043213737826426    | None                    | 0.0                      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.07753528715890391 | 14   |

## cutpoint\_detection tables

*cutpoint\_scan*

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio           | p_value                | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|----------------------|------------------------|---------------------|
| 319      | 0                      | 49               | 3                      | 19               | 0.0                  | 0.02694427434170239    | 1.5695335082504807  |
| 320      | 0                      | 49               | 4                      | 18               | 0.0                  | 0.00752854724253449    | 2.123288820003888   |
| 352      | 0                      | 49               | 5                      | 17               | 0.0                  | 0.0020225947815764307  | 2.6940911176014084  |
| 364      | 0                      | 49               | 13                     | 9                | 0.0                  | 8.560748975173139e-09  | 8.067488237477606   |
| 372      | 0                      | 49               | 14                     | 8                | 0.0                  | 1.3283920823544527e-09 | 8.876673721601218   |
| 381      | 1                      | 48               | 14                     | 8                | 0.011904761904761904 | 1.731570731911155e-08  | 7.761559763513135   |
| 382      | 3                      | 46               | 15                     | 7                | 0.030434782608695653 | 1.0398205008988075e-07 | 6.983041624352472   |
| 400      | 4                      | 45               | 15                     | 7                | 0.04148148148148148  | 4.330878206429079e-07  | 6.36342402938942    |
| 426      | 4                      | 45               | 16                     | 6                | 0.03333333333333333  | 7.233726445978198e-08  | 7.140637918749475   |
| 434      | 5                      | 44               | 16                     | 6                | 0.04261363636363636  | 2.7039639334463545e-07 | 6.567999105485465   |
| 490      | 45                     | 4                | 20                     | 2                | 1.125                | 1.0                    | -0.0                |
| 502      | 46                     | 3                | 20                     | 2                | 1.5333333333333334   | 0.6419888956213136     | 0.19247248378799386 |
| 508      | 47                     | 2                | 20                     | 2                | 2.35                 | 0.5828392348978783     | 0.23445122058408072 |
| 510      | 48                     | 1                | 20                     | 2                | 4.8                  | 0.2249846907532149     | 0.6478470327668787  |
| 511      | 48                     | 1                | 21                     | 1                | 2.2857142857142856   | 0.5267605633802818     | 0.2783867465185951  |
| 521      | 49                     | 0                | 21                     | 1                | None                 | 0.3098591549295775     | 0.508835667896869   |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 20 | 29 |
| 5  | 17 |

*domain\_retention\_descriptives*

| Domain_key      | Domain_name                   | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-----------------|-------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_key_domain | NAB conserved region 2 (NCD2) | 71                      | 49                   | 13              | 9                | 0.35547087642896025                             |

## frequency tables

Partner\_gene\_counts

| Partner_gene | Event_count |
|--------------|-------------|
| CUX2         | 1           |
| DLGAP2       | 1           |
| FAM186A      | 1           |
| FMN1         | 3           |
| KIF5A        | 1           |
| KRT6A        | 1           |
| LFNG         | 1           |
| LGR5         | 1           |
| LRP1         | 3           |
| MGAT4C       | 2           |
| MTOR         | 1           |
| PDZRN4       | 1           |
| STAT6        | 53          |
| TMEM117      | 2           |

Per-event results (results.tsv)

| event_id                        | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_na<br>me  | part<br>ner_<br>gen<br>e | fra<br>me<br>_sta<br>tus | tar<br>get_<br>ro<br>le  | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains | lost<br>_do<br>mai<br>ns                                | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                          | source_site2<br>_effect_on_f<br>rame |
|---------------------------------|---------------------------|----------------------------|----------------------|--------------------------|--------------------------|--------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|--------------------------|---------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0052<br>994-T01-IM<br>6-1 | P-0052<br>994-T0<br>1-IM6 |                            | FAM1<br>86A-<br>NAB2 | FAM<br>186<br>A          | out-<br>of-fr<br>ame     | thr<br>ee_<br>_pr<br>ime | 574883<br>61               | 7                           | 490                                 | True                           | lost                      | 0.0                              | False                       |                          | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                           | FAM186A<br>(NM_001145475) - NAB2<br>(NM_005967)<br>rearrangement: c.193-1572<br>6:FAM186A_c.1469-34:NA<br>B2inv   Protein Fusion: out<br>of frame {FAM186A:NAB2}                | NA                                   |
| EVT-P-0015<br>989-T01-IM<br>6-2 | P-0015<br>989-T0<br>1-IM6 |                            | FMN<br>1-NA<br>B2    | FMN<br>1                 | out-<br>of-fr<br>ame     | thr<br>ee_<br>_pr<br>ime | 574879<br>44               | 7                           | 490                                 | True                           | lost                      | 0.0                              | False                       |                          | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                           | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>Rearrangement : t(12,15)(q<br>13.3,q13.3)(chr12:g.57487<br>944::chr15:g.33270869)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2} | NA                                   |

| event_id                         | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status         | target_role              | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                            | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|--------------------|--------------|----------------------|--------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0015<br>989-T05-IM<br>6-3  | P-0015<br>989-T0<br>5-IM6 |            | FMN<br>1-NA<br>B2  | FMN<br>1     | out-<br>of-fr<br>ame | thr<br>ee_<br>_pr<br>ime | 574879<br>44       | 7               | 490                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>rearrangement: t(12;15)(q1<br>3.3;q13.3)(chr12:g.574879<br>44::chr15:g.33270870)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2}  | NA                           |
| EVT-P-0015<br>989-T02-IM<br>6-4  | P-0015<br>989-T0<br>2-IM6 |            | FMN<br>1-NA<br>B2  | FMN<br>1     | out-<br>of-fr<br>ame | thr<br>ee_<br>_pr<br>ime | 574879<br>45       | 7               | 490                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>Rearrangement : t(12,15)(q<br>13.3,q13.3)(chr12:g.57487<br>944::chr15:g.33270869)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2} | NA                           |
| EVT-P-0020<br>678-T01-IM<br>6-6  | P-0020<br>678-T0<br>1-IM6 |            | KRT6<br>A-NA<br>B2 | KRT<br>6A    | in-fr<br>ame         | thr<br>ee_<br>_pr<br>ime | 574868<br>17       | 5               | 382                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | KRT6A (NM_005554) -<br>NAB2 (NM_005967)<br>rearrangement: c.1204-574<br>:KRT6A_c.1144-29:NAB2in<br>v   Protein Fusion: in frame<br>{KRT6A:NAB2}                                 | NA                           |
| EVT-P-0045<br>531-T01-IM<br>6-7  | P-0045<br>531-T0<br>1-IM6 |            | LGR5<br>-NAB<br>2  | LGR<br>5     | unk<br>now<br>n      | thr<br>ee_<br>_pr<br>ime | 574884<br>58       | 7               | 511                         | False                  | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | LGR5 (NM_003667) -<br>NAB2 (NM_005967)<br>rearrangement: c.285-7692<br>:LGR5_c.1532:NAB2dup  <br>Protein Fusion: mid-exon<br>{LGR5:NAB2}                                        | NA                           |
| EVT-P-0068<br>029-T01-IM<br>7-9  | P-0068<br>029-T0<br>1-IM7 |            | LRP1<br>-NAB<br>2  | LRP<br>1     | unk<br>now<br>n      | thr<br>ee_<br>_pr<br>ime | 574871<br>82       | 6               | 426                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | LRP1 (NM_002332) -<br>NAB2 (NM_005967)<br>rearrangement: c.609:LRP<br>1_c.1277-8:NAB2dup  <br>Protein Fusion: mid-exon<br>{LRP1:NAB2}                                           | NA                           |
| EVT-P-0058<br>847-T01-IM<br>7-10 | P-0058<br>847-T0<br>1-IM7 |            | LRP1<br>-NAB<br>2  | LRP<br>1     | unk<br>now<br>n      | thr<br>ee_<br>_pr<br>ime | 574867<br>22       | 4               | 372                         | False                  | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>) |                   | LRP1 (NM_002332) -<br>NAB2 (NM_005967)<br>rearrangement: c.5662+68<br>5:LRP1_c.1114:NAB2dup  <br>Protein Fusion: mid-exon<br>{LRP1:NAB2}                                        | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name        | part_ner_gene  | fra_me_status        | tar_get_role         | breakp_ooint_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action   | domain_is_trun_cated | retain_ed_do_mains                              | lost_do_mains                                              | disrupt_ed_do_mains                            | source_annotation_text                                                                                                                                                          | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|---------------------|----------------|----------------------|----------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|------------------------------|----------------------|-------------------------------------------------|------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0008<br>384-T01-IM<br>5-12 | P-0008<br>384-T0<br>1-IM5 |              | NAB2<br>-CUX<br>2   | CUX<br>2       | out-<br>of-fr<br>ame | thr_ee<br>_pr<br>ime | 574864<br>06          | 3                 | 364                          | True                     | disru<br>pted  | 0.0179640<br>718562874<br>25 | True                 |                                                 |                                                            | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | CUX2 (NM_015267) -<br>NAB2 (NM_005967)<br>rearrangement: c.63+8305<br>8:CUX2_c.1091+42:NAB2<br>dup   Protein fusion: out of<br>frame (CUX2-NAB2)                                | NA                            |
| EVT-P-0038<br>472-T01-IM<br>6-13 | P-0038<br>472-T0<br>1-IM6 |              | NAB2<br>-DLG<br>AP2 | DLG<br>AP2     | out-<br>of-fr<br>ame | five<br>_pr<br>ime   | 574875<br>47          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                          | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |                                                            |                                                | NAB2 (NM_005967) -<br>DLGAP2 (NM_004745)<br>rearrangement: t(8;12)(p23<br>.3;q13.3)(chr8:g.1518542::<br>chr12:g.57487547)  <br>Protein Fusion: out of<br>frame {NAB2:DLGAP2}    | NA                            |
| EVT-P-0039<br>139-T01-IM<br>6-14 | P-0039<br>139-T0<br>1-IM6 |              | NAB2<br>-KIF5<br>A  | KIF5<br>A      | in-fr<br>ame         | five<br>_pr<br>ime   | 574860<br>74          | 3                 | 320                          | True                     | disru<br>pted  | 0.7245508<br>982035929       | True                 |                                                 |                                                            | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>KIF5A (NM_004984)<br>rearrangement: c.958-157:<br>NAB2_c.129+5819:KIF5Ad<br>el   Protein Fusion: in frame<br>{NAB2:KIF5A}                                 | NA                            |
| EVT-P-0022<br>192-T01-IM<br>6-18 | P-0022<br>192-T0<br>1-IM6 |              | NAB2<br>-LRP<br>1   | LRP<br>1       | out-<br>of-fr<br>ame | thr_ee<br>_pr<br>ime | 574865<br>54          | 4                 | 364                          | True                     | disru<br>pted  | 0.0179640<br>718562874<br>25 | True                 |                                                 |                                                            | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | LRP1 (NM_002332) -<br>NAB2 (NM_005967)<br>rearrangement: c.842-2141<br>:LRP1_c.1092-146:NAB2d<br>up   Protein Fusion: out of<br>frame {LRP1:NAB2}                               | NA                            |
| EVT-P-0028<br>347-T01-IM<br>6-21 | P-0028<br>347-T0<br>1-IM6 |              | NAB2<br>-MGA<br>T4C | MG<br>AT4<br>C | unk<br>now<br>n      | five<br>_pr<br>ime   | 574863<br>29          | 3                 | 352                          | False                    | disru<br>pted  | 0.9161676<br>646706587       | True                 |                                                 |                                                            | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967)<br>rearrangement: c.1056:NA<br>B2_chr12:g.86871854inv  <br>Protein Fusion: mid-exon<br>{NAB2:MGAT4C}                                                           | NA                            |
| EVT-P-0011<br>161-T01-IM<br>5-22 | P-0011<br>161-T0<br>1-IM5 |              | NAB2<br>-MGA<br>T4C | MG<br>AT4<br>C | unk<br>now<br>n      | thr_ee<br>_pr<br>ime | 574865<br>34          | 4                 | 364                          | True                     | disru<br>pted  | 0.0179640<br>718562874<br>25 | True                 |                                                 |                                                            | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967)<br>rearrangement: c.1092-166<br>:NAB2_chr12:g.87188994d<br>el   Antisense fusion                                                                               | NA                            |
| EVT-P-0025<br>245-T01-IM<br>6-23 | P-0025<br>245-T0<br>1-IM6 |              | NAB2<br>-MTO<br>R   | MT<br>OR       | unk<br>now<br>n      | thr_ee<br>_pr<br>ime | 574919<br>19          | 7                 | 526                          | True                     | lost           | 0.0                          | False                |                                                 | NAB<br>cons<br>erve<br>d<br>re<br>gion<br>2 (N<br>CD2<br>) |                                                | MTOR (NM_004958) -<br>STAT6 (NM_003153)<br>rearrangement: t(1;12)(p36<br>.22;q13.3)(chr1:g.1130031<br>0::chr12:g.57491919)  <br>Transcript Fusion<br>{MTOR:NAB2}                | NA                            |
| EVT-P-0001<br>055-T02-IM<br>5-49 | P-0001<br>055-T0<br>2-IM5 |              | NAB2<br>-STA<br>T6  | STA<br>T6      | unk<br>now<br>n      | five<br>_pr<br>ime   | 574879<br>38          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                          | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |                                                            |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exon 6 fused<br>with STAT6 exon16) : c.14<br>69-457:NAB2_c.1760:STA<br>T6inv   Protein fusion:<br>mid-exon (NAB2-STAT6) | NA                            |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name       | part_ner_gene | fra_me_status   | tar_get_role       | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains                              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|--------------------|---------------|-----------------|--------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------------------------------------|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0066<br>534-T01-IM<br>7-50 | P-0066<br>534-T0<br>1-IM7 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574880<br>34          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1469-361<br>:NAB2_c.1775:STAT6inv  <br>Protein Fusion: mid-exon<br>{NAB2:STAT6}                                                            | NA                            |
| EVT-P-0003<br>845-T01-IM<br>5-51 | P-0003<br>845-T0<br>1-IM5 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574878<br>34          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967)<br>-STAT6 (NM_003153)<br>fusion (NAB2 exon 6 fused<br>to STAT6 exon 16) : c.146<br>8+453:NAB2_c.1744+30:S<br>TAT6inv   Protein fusion: in<br>frame (NAB2-STAT6)                       | NA                            |
| EVT-P-0029<br>101-T01-IM<br>6-52 | P-0029<br>101-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574868<br>24          | 5                 | 382                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>fused to STAT6 exons<br>16-22): c.1144-22:NAB2_c.<br>-1269:STAT6inv  <br>Transcript Fusion<br>{NAB2:STAT6}                        | NA                            |
| EVT-P-0029<br>763-T01-IM<br>6-53 | P-0029<br>763-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574875<br>87          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>fused to STAT6 exons<br>15-22): c.1468+206:NAB2_<br>c.1624:STAT6inv   Protein<br>Fusion: mid-exon<br>{NAB2:STAT6}                 | NA                            |
| EVT-P-0001<br>055-T03-IM<br>6-55 | P-0001<br>055-T0<br>3-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574879<br>38          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1469-457<br>:NAB2_c.1760:STAT6inv  <br>Protein Fusion: mid-exon<br>{NAB2:STAT6}                                                            | NA                            |
| EVT-P-0020<br>523-T01-IM<br>6-56 | P-0020<br>523-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574875<br>05          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>fused in frame with STAT6<br>exons 16-22): c.1468+124:<br>NAB2_c.1745-112:STAT6i<br>nv   Protein Fusion: in<br>frame {NAB2:STAT6} | NA                            |
| EVT-P-0032<br>069-T01-IM<br>6-57 | P-0032<br>069-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574877<br>12          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                     | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>fused to STAT6 exons<br>16-22): c.1468+331:NAB2_<br>c.1870:STAT6inv   Protein<br>Fusion: mid-exon<br>{NAB2:STAT6}                 | NA                            |



| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains               | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022<br>259-T02-IM<br>6-61 | P-0022<br>259-T0<br>2-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488455           | 7               | 510                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-7 fused with STAT6 exons 16-22): c.1529:NAB2_c.1745-128:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}              | NA                           |
| EVT-P-0004<br>282-T03-IM<br>5-62 | P-0004<br>282-T0<br>3-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488115           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exon 16 to 22) : c.1469-280:NAB2_c.1745-157:STAT6inv   Protein fusion: in frame (NAB2-STAT6)    | NA                           |
| EVT-P-0033<br>539-T02-IM<br>6-63 | P-0033<br>539-T0<br>2-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57486902           | 5               | 400                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) rearrangement: c.1200:NA B2_chr12:g.57486902inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                                        | NA                           |
| EVT-P-0019<br>357-T03-IM<br>6-64 | P-0019<br>357-T0<br>3-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488027           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-368:NAB2_c.1690:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                | NA                           |
| EVT-P-0019<br>182-T01-IM<br>6-66 | P-0019<br>182-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487665           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+284: NAB2_c.1744+39:STAT6inv   Protein Fusion: in frame {NAB2:STAT6} | NA                           |
| EVT-P-0036<br>288-T01-IM<br>6-67 | P-0036<br>288-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488128           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1469-267:NAB2_c.1850:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                            | NA                           |
| EVT-P-0036<br>975-T01-IM<br>6-68 | P-0036<br>975-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57486781           | 4               | 381                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) rearrangement: c.1143+30: NAB2_chr12:g.57508083inv   Transcript Fusion {NAB2:STAT6}                                                                            | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains              | lost_domains | disrupted_domains             | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------|--------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0019<br>032-T01-IM<br>6-69 | P-0019<br>032-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488347           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-48:NAB2_c.1691:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}      | NA                           |
| EVT-P-0017<br>518-T01-IM<br>6-70 | P-0017<br>518-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57486597           | 4               | 364                         | True                   | disrupted     | 0.9880239520958084       | True                |                               |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-2 fused to STAT6 exons 18-22): c.1092-103:NAB2_c.2042:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}     | NA                           |
| EVT-P-0016<br>751-T02-IM<br>6-71 | P-0016<br>751-T0<br>2-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487958           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1469-437:NAB2_c.1745-109:STAT6inv   Protein Fusion: in frame {NAB2:STAT6} | NA                           |
| EVT-P-0038<br>591-T01-IM<br>6-72 | P-0038<br>591-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487747           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon6 fused with STAT6 exon16) : c.1468+366:NAB2_c.1745-67:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}        | NA                           |
| EVT-P-0016<br>113-T01-IM<br>6-73 | P-0016<br>113-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57486482           | 3               | 364                         | True                   | disrupted     | 0.9880239520958084       | True                |                               |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon3 fused to STAT6 exon 18: c.1091+118:NAB2_c.2051:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}              | NA                           |
| EVT-P-0004<br>405-T01-IM<br>5-74 | P-0004<br>405-T0<br>1-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487215           | 6               | 434                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) -STAT6(NM_003153) Fusion ( NAB2 exon 6 fused to STAT6 exon 18) : c.1302:NAB2_c.1969:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6)                | NA                           |
| EVT-P-0004<br>943-T01-IM<br>5-75 | P-0004<br>943-T0<br>1-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488487           | 7               | 521                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-7 with STAT6 exons 15-22) : c.1561:NAB2_c.1739:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains               | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0040<br>364-T01-IM<br>6-76 | P-0040<br>364-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487738           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0040<br>364-T02-IM<br>6-77 | P-0040<br>364-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487738           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0014<br>386-T01-IM<br>6-78 | P-0014<br>386-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487858           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 17 - 22): c.1468+477:NAB2_c.1891+67:STAT6inv   Protein fusion: in frame (NAB2-STAT6) | NA                           |
| EVT-P-0043<br>579-T01-IM<br>6-79 | P-0043<br>579-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57488047           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-348:NAB2_c.1744+96:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0044<br>036-T01-IM<br>6-80 | P-0044<br>036-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57488037           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0044<br>232-T02-IM<br>6-81 | P-0044<br>232-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488269           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                           | NA                           |
| EVT-P-0001<br>947-T03-IM<br>6-82 | P-0001<br>947-T0<br>3-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487383           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                             | NA                           |
| EVT-P-0045<br>590-T01-IM<br>6-83 | P-0045<br>590-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487780           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                           | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains              | lost_domains | disrupted_domains             | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------|--------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0046<br>398-T01-IM<br>6-84 | P-0046<br>398-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487473           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0046<br>398-T02-IM<br>6-85 | P-0046<br>398-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487473           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0048<br>656-T01-IM<br>6-86 | P-0048<br>656-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57486417           | 3               | 364                         | True                   | disrupted     | 0.9880239520958084       | True                |                               |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1091+53: NAB2_c.1956-25:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                           | NA                           |
| EVT-P-0013<br>489-T01-IM<br>5-87 | P-0013<br>489-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487506           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 15 to 22) : c.1468+125:NAB2_c.1625:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |
| EVT-P-0053<br>200-T02-IM<br>6-89 | P-0053<br>200-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487609           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+228:NAB2_c.1891+45:STAT6 inv   Protein Fusion: in frame {NAB2:STAT6}                                          | NA                           |
| EVT-P-0013<br>261-T01-IM<br>5-90 | P-0013<br>261-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487567           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 15): c.1468+186:NAB2_c.1703:STAT6 inv   Protein fusion: mid-exon (NAB2-STAT6)               | NA                           |
| EVT-P-0013<br>216-T02-IM<br>5-91 | P-0013<br>216-T0<br>2-IM5 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487806           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1468+425:NAB2_c.1745-142:STAT6inv   Protein fusion: in frame (NAB2-STAT6)                                                 | NA                           |
| EVT-P-0055<br>865-T01-IM<br>6-92 | P-0055<br>865-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57488325           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2) |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-70: NAB2_c.1744+36:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                           | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name       | part_ner_gene | fra_me_status   | tar_get_role       | breakp_ooint_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains                              | lost_do_mains | disrupt_ed_do_mains                            | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|--------------------|---------------|-----------------|--------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------------------------------------|---------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0055<br>909-T01-IM<br>6-93  | P-0055<br>909-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574859<br>66          | 2                 | 319                          | True                     | disru<br>pted  | 0.7185628<br>74251497      | True                 |                                                 |               | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.957+185:<br>NAB2_c.340-42:STAT6inv<br>  Protein Fusion: in frame<br>{NAB2:STAT6}                                                           | NA                            |
| EVT-P-0012<br>841-T01-IM<br>5-94  | P-0012<br>841-T0<br>1-IM5 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574876<br>86          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>with STAT6 exons 15-22) :<br>c.1468+305:NAB2_c.1712:<br>STAT6inv   Protein fusion:<br>mid-exon (NAB2-STAT6)                       | NA                            |
| EVT-P-0057<br>770-T01-IM<br>6-95  | P-0057<br>770-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574874<br>67          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1468+86:<br>NAB2_c.1744+3:STAT6inv<br>  Protein Fusion: in frame<br>{NAB2:STAT6}                                                           | NA                            |
| EVT-P-0057<br>786-T01-IM<br>6-96  | P-0057<br>786-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574882<br>44          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1469-151<br>:NAB2_c.1892-23:STAT6in<br>v   Protein Fusion: in frame<br>{NAB2:STAT6}                                                        | NA                            |
| EVT-P-0012<br>421-T01-IM<br>5-97  | P-0012<br>421-T0<br>1-IM5 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574865<br>09          | 3                 | 364                          | True                     | disru<br>pted  | 0.9880239<br>520958084     | True                 |                                                 |               | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-3<br>fused in-frame with STAT6<br>exons 20-22): c.1091+145:<br>NAB2_c.2159+239:STAT6i<br>nv   Protein fusion: in frame<br>(NAB2-STAT6) | NA                            |
| EVT-P-0007<br>527-T01-IM<br>5-98  | P-0007<br>527-T0<br>1-IM5 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574881<br>11          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>reciprocal fusion (NAB2<br>exons 1-6 with STAT6<br>exons 16-22) : c.1469-284:<br>NAB2_c.1745-161:STAT6i<br>nv   Protein fusion: in frame<br>(NAB2-STAT6)    | NA                            |
| EVT-P-0001<br>055-T04-IM<br>6-99  | P-0001<br>055-T0<br>4-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574879<br>38          | 7                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1469-457<br>:NAB2_c.1760:STAT6inv  <br>Protein Fusion: mid-exon<br>{NAB2:STAT6}                                                            | NA                            |
| EVT-P-0059<br>117-T01-IM<br>7-100 | P-0059<br>117-T0<br>1-IM7 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574884<br>48          | 7                 | 508                          | False                    | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2) |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>rearrangement: c.1522:NA<br>B2_c.287:STAT6inv  <br>Protein Fusion: mid-exon<br>{NAB2:STAT6}                                                                 | NA                            |

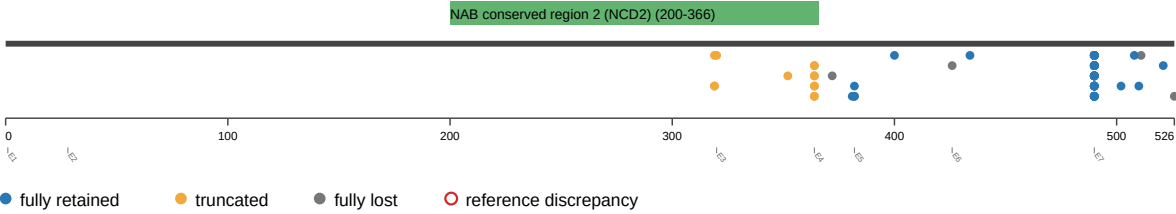
| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains               | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0059<br>275-T01-IM<br>7-101 | P-0059<br>275-T0<br>1-IM7 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487481           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+100:NAB2_c.1801:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0006<br>897-T01-IM<br>5-102 | P-0006<br>897-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488251           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2(NM_005967) - STAT6(NM_003153) Fusion ( NAB2 exons 1 to 6 fused with STAT6 exons 16 to 22) : c.1469-144:NAB2_c.1811:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |
| EVT-P-0006<br>243-T01-IM<br>5-103 | P-0006<br>243-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488431           | 7               | 502                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2(NM_005967) -STAT6(NM_003153) Fusion ( NAB2 exons 1 to 7 fused with STAT6 exon 16 to 22) : c.1505:NAB2_c.1745-142:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6)   | NA                           |
| EVT-P-0005<br>578-T01-IM<br>5-104 | P-0005<br>578-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57486815           | 5               | 382                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon4 fused with STAT6 exon5) : c.1144-31:NAB2_c.340-81:STAT6inv   Protein fusion: in frame (NAB2-STAT6)               | NA                           |
| EVT-P-0067<br>841-T01-IM<br>7-106 | P-0067<br>841-T0<br>1-IM7 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487573           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0066<br>305-T01-IM<br>7-107 | P-0066<br>305-T0<br>1-IM7 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57488138           | 7               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-257:NAB2_c.1745-14:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                           | NA                           |
| EVT-P-0027<br>038-T01-IM<br>6-108 | P-0027<br>038-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487452           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2) |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name              | partner_gene    | frame_status         | target_role             | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction     | domain_is_truncated | retained_domains | lost_domains | disrupted_domains                              | source_annotation_text                                                                                                                                            | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------------------|-----------------|----------------------|-------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|------------------------------|---------------------|------------------|--------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0030<br>371-T02-IM<br>6-109 | P-0030<br>371-T0<br>2-IM6 |            | NAB2<br>-TME<br>M117     | TME<br>M11<br>7 | out-<br>of-fr<br>ame | five<br>_pr<br>ime      | 574858<br>85       | 2               | 319                         | True                   | disru<br>pted | 0.7185628<br>74251497        | True                |                  |              | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>TMEM117 (NM_032256)<br>rearrangement: c.957+104:<br>NAB2_c.278-424:TMEM11<br>7dup   Protein Fusion: out<br>of frame {NAB2:TMEM117}          | NA                           |
| EVT-P-0004<br>638-T01-IM<br>5-110 | P-0004<br>638-T0<br>1-IM5 |            | PDZ<br>RN4-<br>NAB2      | PDZ<br>RN4      | out-<br>of-fr<br>ame | thr<br>ee<br>_pr<br>ime | 574866<br>31       | 4               | 364                         | True                   | disru<br>pted | 0.0179640<br>718562874<br>25 | True                |                  |              | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | PDZRN4 (NM_001164595)<br>- NAB2 (NM_005967) -<br>rearrangement: c.843+122<br>859:PDZRN4_c.1092-69:N<br>AB2del.   Protein fusion:<br>out of frame<br>(PDZRN4-NAB2) | NA                           |
| EVT-P-0030<br>371-T01-IM<br>6-111 | P-0030<br>371-T0<br>1-IM6 |            | TME<br>M117<br>-NAB<br>2 | TME<br>M11<br>7 | out-<br>of-fr<br>ame | five<br>_pr<br>ime      | 574858<br>85       | 2               | 319                         | True                   | disru<br>pted | 0.7185628<br>74251497        | True                |                  |              | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>TMEM117 (NM_032256)<br>Rearrangement : c.957+10<br>4:NAB2_c.278-424:TMEM<br>117dup   Protein Fusion:<br>out of frame<br>{NAB2:TMEM117}      | NA                           |

# Figures

## domain retention outliers

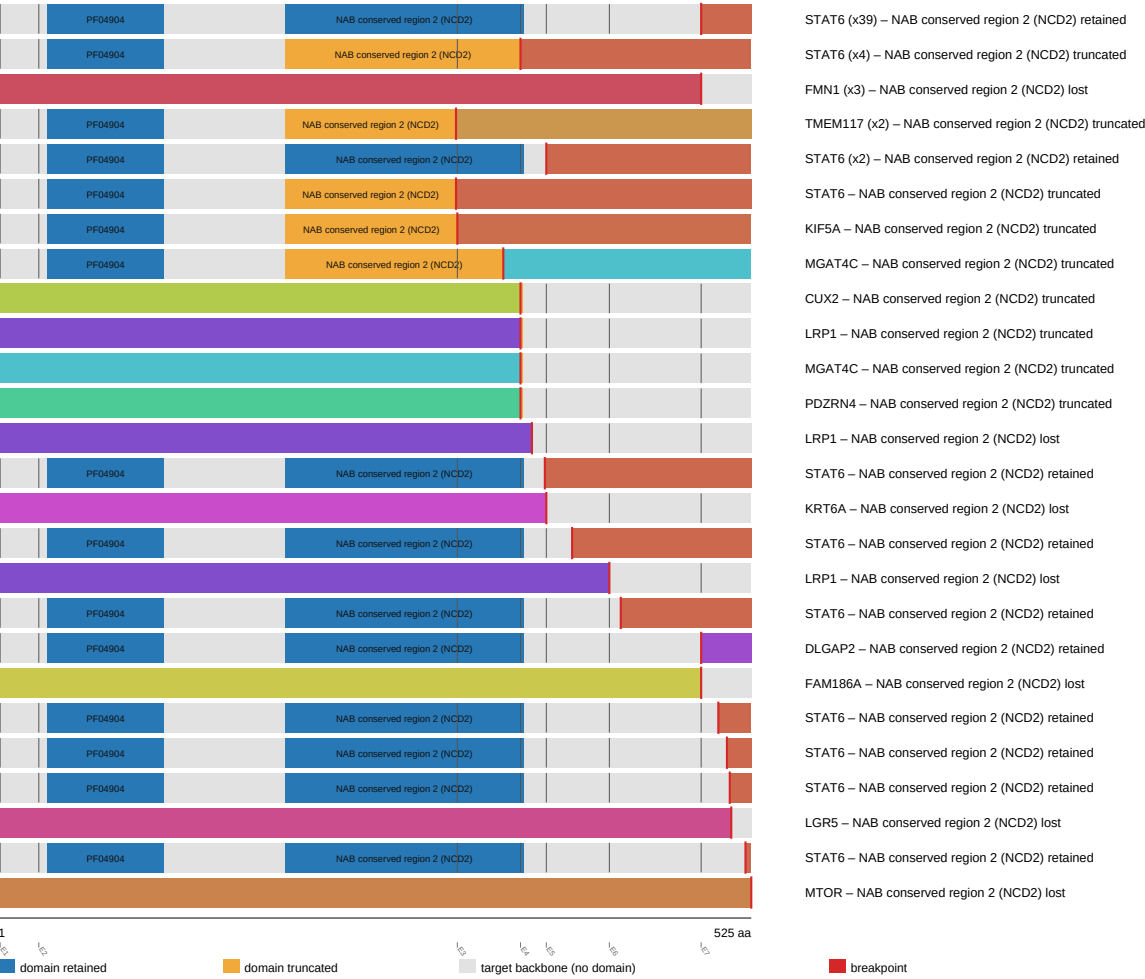
NAB2 domain-retention track



## fusion schematic

NAB2 fusion-transcript schematic

partner block → breakpoint → retained NAB2 portion (domain-colored, exon ticks)



## intragenic deletion schematic



NAB2 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk\_impact\_50k\_2026

